

MEMORANDUM

To: VP, Category Management
From: Nilkanth Changawala, Data Analyst
Date: August 17, 2026
Re: Promotion discount depth, Eggs and Bath Tissues

Bottom line

We are very likely over discounting on three of four active promotion programs across Eggs and Bath Tissues, giving away margin without a matching volume payoff. One program, Eggs mailer promotions, appears under discounted relative to its revenue maximizing depth. Before changing any budget, we recommend a four week controlled test to confirm these patterns hold outside of historical, non randomized promotion data.

Background

Using one year of household level transaction data covering 2,469 households, we built a price elasticity and promotion response model for two staple grocery categories, correcting for the fact that price and promotion are not randomly assigned by category managers, which otherwise biases a naive analysis.

What we found

- Eggs and Bath Tissues are both price inelastic, as expected for routine grocery staples, but Eggs is meaningfully more price sensitive than a simple, uncorrected analysis would suggest.
- In store display and mailer promotions work differently by category. Eggs responds strongly to both. Bath Tissues responds weakly and inconsistently.
- Discount depth, how big the price cut is during a promotion, does not uniformly help. For Eggs display, Bath Tissues display, and Bath Tissues mailer promotions, our model finds the revenue maximizing discount is close to zero, while current practice runs 20 to 28 percent. For Eggs mailer promotions, the opposite holds, current practice of about 26 percent sits below the roughly 40 percent depth our model estimates as revenue maximizing.

Category	Promo type	Current avg. discount	Model optimal discount
Eggs	Mailer	26%	40%
Eggs	Display	21%	0%
Bath Tissues	Display	24%	0%
Bath Tissues	Mailer	28%	0%

Estimated opportunity, and how much to trust it

Modeled at roughly 31 percent of the revenue these four programs generate, concentrated mostly in the three "reduce discount toward zero" recommendations. Those three are corner solutions, the least trustworthy kind of model output, since they push to the edge of the data and rely most heavily on the assumption that discount depth is unrelated to demand once we control for product, store, and week. The fourth recommendation, deepen the Eggs mailer discount, is an interior optimum found well within the range of discounts already observed, and warrants more confidence than the other three.

Recommendation

Run a four week controlled test. Randomly assign discount depth, not category manager judgment, for Eggs and Bath Tissues promotions across a sample of comparable stores. Confirm the direction and rough size of the effect before committing budget at scale. If confirmed even at a fraction of the modeled size, this remains a meaningful margin opportunity across just these two categories, and the same method extends to any category once the pilot validates the approach.

Full analysis, model specification, and every caveat are documented in the project repository.